

1. Вошел под пользователем user1

```
root@eltex-practice2-pg1-v13:~# su user1
bash: /home/user1/.bashrc: Permission denied
user1@eltex-practice2-pg1-v13:/root$
```

2. С помощью команды ps и фильтрации awk подсчитали количество процессов имеющих несколько потоков

```
6
user1@eltex-practice2-pg1-v13:/root$ ps -eo nlwp | awk '$1 > 1 {print $1}' | wc -l
10
user1@eltex-practice2-pg1-v13:/root$
```

3. В интерактивном режиме top изменил вывод команды

```
Fields Management for window 1:Def, whose current sort field is %CPU
Navigate with Up/Dn, Right selects for move then <Enter> or Left commits,
'd' or <Space> toggles display, 's' sets sort. Use 'q' or <Esc> to end!

* PID      = Process Id          nsIPC      = IPC namespace Inode
* USER     = Effective User Name nsMNT      = MNT namespace Inode
* RUSER    = Real User Name     nsNET      = NET namespace Inode
* PR       = Priority           nsPID      = PID namespace Inode
* NI       = Nice Value         nsUSER     = USER namespace Inode
* VIRT     = Virtual Image (KiB) nsUTS      = UTS namespace Inode
RES       = Resident Size (KiB) LXC         = LXC container name
SHR       = Shared Memory (KiB) RSan       = RES Anonymous (KiB)
* S        = Process Status     RSfd       = RES File-based (KiB)
* %CPU     = CPU Usage          RSlk       = RES Locked (KiB)
* %MEM     = Memory Usage (RES) RSsh       = RES Shared (KiB)
* TIME+   = CPU Time, hundredths CGNAME     = Control Group name
* COMMAND  = Command Name/Line  NU         = Last Used NUMA node
PPID      = Parent Process pid  LOGID      = Login User Id
UID       = Effective User Id   EXE        = Executable Path
```

Результат выполнения

PID USER RUSER PR NI S %CPU %MEM TIME+ COMMAND									
679	root	root	20	0	S	0.1	0.1	9:06.69	qemu-ga
39950	root	root	20	0	I	0.0	0.0	0:00.82	kworker/u4:1-flush-8:0
40270	root	root	20	0	I	0.0	0.0	0:00.43	kworker/0:1-events
40308	user1	user1	20	0	R	0.0	0.2	0:00.03	top
359	root	root	rt	0	S	0.0	0.7	1:22.50	multipathd
39996	root	root	20	0	I	0.0	0.0	0:00.66	kworker/u4:2-events_unbound
40078	root	root	20	0	S	0.0	0.3	0:00.23	sshd
40240	root	root	20	0	I	0.0	0.0	0:00.17	kworker/u4:4-events_unbound
40279	root	root	20	0	I	0.0	0.0	0:00.11	kworker/1:3-events_freezable
1	root	root	20	0	S	0.0	0.4	0:25.73	systemd
2	root	root	20	0	S	0.0	0.0	0:00.30	kthreadd
3	root	root	20	0	S	0.0	0.0	0:00.00	pool_workqueue_release

5. все процессы вместе с командой passwd, для которых реальным пользователем является пользователь, которым вошли в сеанс;

PID	USER	RUSER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
40707	user1	user1	20	0	S	0.3	0.2	4:44.12	top
44645	user1	user1	20	0	R	0.3	0.2	0:00.14	top
40314	user1	user1	20	0	S	0.0	0.1	0:00.02	screen
40315	user1	user1	20	0	S	0.0	0.1	0:00.01	bash
40332	user1	user1	20	0	S	0.0	0.1	0:00.60	screen
40333	user1	user1	20	0	S	0.0	0.1	0:00.01	screen
40334	user1	user1	20	0	S	0.0	0.1	0:00.00	bash
40436	user1	user1	20	0	S	0.0	0.1	0:00.01	screen
40437	user1	user1	20	0	S	0.0	0.1	0:00.00	bash
40675	user1	user1	20	0	S	0.0	0.1	0:06.67	screen
40676	user1	user1	20	0	S	0.0	0.1	0:00.00	bash
40693	user1	user1	20	0	S	0.0	0.1	0:00.01	bash
40697	user1	user1	20	0	S	0.0	0.1	0:00.00	bash
40713	user1	user1	20	0	S	0.0	0.1	0:00.00	bash
40716	root	user1	20	0	S	0.0	0.1	0:00.00	passwd

Поочередно запустил сигналы 15,2,3,9 на 9 сигнале процесс завершился

```
top - 14:15:24 up 8 days, 12:45, 6 users, load average: 0.00, 0.00
Tasks: 145 total, 1 running, 144 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.3 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 st
MiB Mem : 3915.9 total, 2225.0 free, 540.0 used, 1440.0 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used, 3375.9 available
Send pid 40716 signal [15/sigterm] 15
```

PID	USER	RUSER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
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Завершение можно увидеть на втором окне

```
user1@eltex-practice2-pg1-v13:/$ passwd
Changing password for user1.
Current password: Killed
```

7.

```
user1@eltex-practice2-pg1-v13:/root$ jobs
[1]-  Stopped                  vim ~/file_task3.txt
[2]+  Stopped                  sleep 600
```

8.

```
user1@eltex-practice2-pg1-v13:/root$ bg %2
[2]+ sleep 600 &
```

9.

```
user1@eltex-practice2-pg1-v13:/root$ renice 10 44861
44861 (process ID) old priority 0, new priority 10
```

Значение приоритета изменился на 10 (3 столбец)

44861	user1	30	10	5684	2168	2060	S	0.0	0.1	0:00.00	sleep
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10.

```
user1@eltex-practice2-pg1-v13:/root$ fg %1
vim ~/file_task3.txt
[2] Done sleep 600
```

11.

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
44897	user1	20	0	11952	6048	3804	R	0.3	0.2	0:00.10	top
40314	user1	20	0	8952	3232	2132	S	0.0	0.1	0:00.02	screen
40315	user1	20	0	7604	4444	3816	S	0.0	0.1	0:00.01	bash
40332	user1	20	0	8292	4108	3808	S	0.0	0.1	0:00.61	screen
40333	user1	20	0	8584	2900	2132	S	0.0	0.1	0:00.01	screen
40334	user1	20	0	7604	4456	3828	S	0.0	0.1	0:00.00	bash
40436	user1	20	0	9216	3528	2132	S	0.0	0.1	0:00.01	screen
40437	user1	20	0	7604	4452	3828	S	0.0	0.1	0:00.00	bash
40675	user1	20	0	9816	4144	2132	S	0.0	0.1	0:06.89	screen
40676	user1	20	0	7604	4452	3832	S	0.0	0.1	0:00.00	bash
40693	user1	20	0	7604	4496	3868	S	0.0	0.1	0:00.01	bash
40697	user1	20	0	7604	4460	3836	S	0.0	0.1	0:00.00	bash
40707	user1	20	0	11944	6056	3812	S	0.0	0.2	4:51.86	top
40713	user1	20	0	7604	4484	3864	S	0.0	0.1	0:00.00	bash
40739	user1	20	0	8952	3288	2132	S	0.0	0.1	0:00.44	screen
40740	user1	20	0	7604	4456	3836	S	0.0	0.1	0:00.00	bash
40744	user1	20	0	7604	4436	3808	S	0.0	0.1	0:00.00	bash
44428	user1	20	0	9512	3836	2132	S	0.0	0.1	0:00.69	screen
44433	user1	20	0	7604	4500	3872	S	0.0	0.1	0:00.01	bash
44487	user1	20	0	7604	4444	3816	S	0.0	0.1	0:00.00	bash
44491	user1	20	0	8292	4096	3784	S	0.0	0.1	0:00.02	screen
44497	user1	20	0	8292	4084	3784	S	0.0	0.1	0:00.02	screen
44498	user1	20	0	8584	2936	2132	S	0.0	0.1	0:00.41	screen
44499	user1	20	0	7604	4440	3816	S	0.0	0.1	0:00.00	bash
44503	user1	20	0	8292	4084	3784	S	0.0	0.1	0:00.01	screen
44504	user1	20	0	8584	2932	2132	S	0.0	0.1	0:00.42	screen
44505	user1	20	0	7604	4500	3872	S	0.0	0.1	0:00.02	bash
44518	user1	20	0	7604	4460	3836	S	0.0	0.1	0:00.01	bash
44760	user1	20	0	7604	4452	3832	S	0.0	0.1	0:00.00	bash
44762	user1	20	0	7604	4488	3872	S	0.0	0.1	0:00.00	bash
44835	user1	20	0	7604	4460	3832	S	0.0	0.1	0:00.03	bash
44876	user1	20	0	24680	13328	8024	T	0.0	0.3	0:00.03	vim

Отправил сигнал 15 процессу sleep, после которого он пропал

```
[5]- Terminated sleep 600
```

12.

```
trap 'echo "Меня голыми руками не возьмешь!"' SIGINT SIGQUIT
```

Проверили работоспособность

```
Меня голыми руками не возьмешь!root$ ^C
Меня голыми руками не возьмешь!root$ ^\
```